UNIT 2 · 9 hours

Concept of Rotating Magnetic Field



Transformers and Electrical Machines

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Explain how three-phase currents set up a rotating field
- Show that the resultant flux has constant magnitude 1.5φm
- Calculate synchronous speed
- State how to reverse the direction of rotation

Syllabus context

Three-phase induction motor: concept of rotating magnetic field, principle of operation, constructional features of motor, types: squirrel cage and wound rotor.

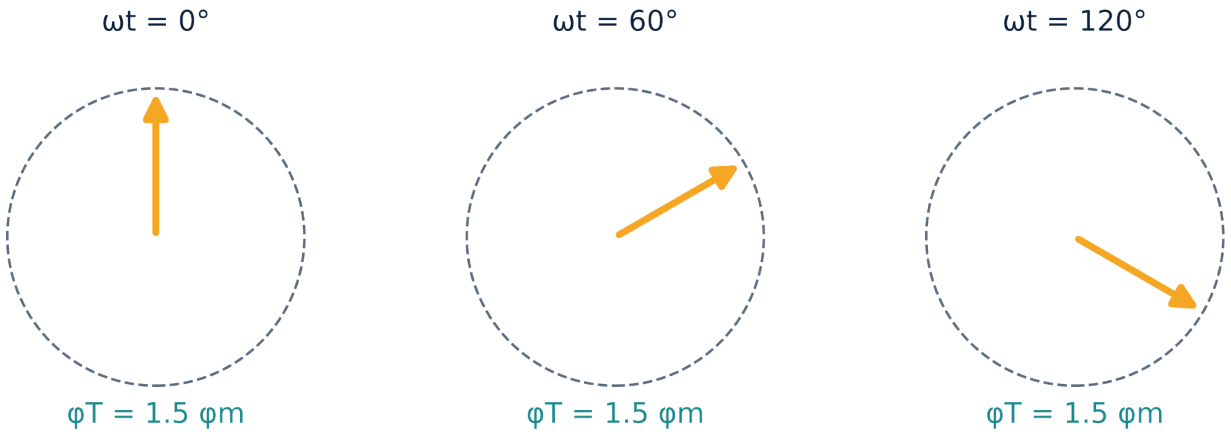
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Stator winding and three fluxes

- Three identical coils placed 120° apart in space.
- Fed from a balanced three-phase supply (120° apart in time).
- $\varphi R = \varphi m \sin \omega t$, $\varphi Y = \varphi m \sin(\omega t - 120°)$, $\varphi B = \varphi m \sin(\omega t - 240°)$.
- Each flux is pulsating along its own axis; their vector sum is what matters.

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Resultant flux at successive instants



- At ωt = 0: φR = 0, φY = −0.866φm, φB = +0.866φm → φT = 1.5φm.
- At 60°: resultant has turned 60°, magnitude still 1.5φm.
- At 120°: turned 120°.
- After 360° it completes one revolution (for 2 poles).

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Synchronous speed

$$N_s = 120\,f / P$$

rpm

$$\varphi_T = 1.5\,\varphi_m$$

constant amplitude

- f = supply frequency (Hz), P = number of poles.
- 4 poles, 50 Hz: Ns = 1500 rpm; 2 poles: 3000 rpm; 6 poles: 1000 rpm; 8 poles: 750 rpm.
- Direction reverses if any two supply leads are interchanged (phase sequence reversed).

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Synchronous speeds at 50 Hz

Poles P	2	4	6	8	10
Ns (rpm)	3000	1500	1000	750	600

Ns = 120 f / P is independent of the load.

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Why it matters

- The rotating field replaces the mechanical motion of a magnet.
- A conductor in this field sees relative motion and has an EMF induced.
- No slip rings or commutator are needed on the stator.
- Makes the three-phase induction motor self-starting.

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Key takeaways

- Three coils 120° apart + three-phase supply → rotating field
- Resultant magnitude = 1.5 φm, speed Ns = 120f/P
- Interchanging two leads reverses rotation
- The rotating field makes the induction motor self-starting

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Practice questions

Q1. Show that the resultant of three fluxes is 1.5φm at ωt = 60°.

Q2. Find the synchronous speed of a 6-pole, 50 Hz motor.

Q3. How do you reverse the direction of an induction motor?

Q4. Why is a single-phase stator winding not self-starting?